Problem 1) Definition of a robot: A robot is defined as intelligence embodied in an engineered construct, with the ability to process information, sense, plan and move within or substantially alter its working environment.

a) ChatGPT is not a robot. While it does have an embodied sense of intelligence, it lacks a physical component or an engineered construct to exist in the physical world. Thus it is not a robot.
b) Washing machine: while it is programmable and is a physical construct, it lacks autonomy and decision making, thus it is not a robot.
c) A remote controlled car is not a robot because it requires human control, not autonomy.
d) Yes as it operates autonomously and uses sensors and algorithms to track where the booster is rather than human control. Is also programmable.
e) It lacks the physical aspect because it can't move within its environment even though it is autonomous.

Problem 2)

a) Active sensors use some sort of waves or actively interact with their environment to get data. For example LiDAR which sends out waves to get data would be an active sensor. A passive sensor simply just measures data in its environment, like a seismic sensor. Proprioceptive sensors give info about internal information of the robot, like RPM or internal temperature. Exteroceptive sensors take in info on the environment, allowing robots to make appropriate decisions.
b) GPS: passive exteroceptive, takes in satellite signals related to environment around it
Bumper Switch: passive exteroceptive, detects physical contact and senses collision w/ environment
Accelerometer: passive proprioceptive, takes in info about device's own acceleration
LiDAR: active exteroceptive, sends out waves to gather info about environment around it
Infrared Proximity Sensor: active exteroceptive, emits infrared light to detect reflects on its environment
Wheel Encoder: passive proprioceptive, tracks wheel movement of internal robot

Problem 3)

a) Because it is unable to detect changes in terrain around it and area in more than a 30º cone. One example could be if a wall appears behind the robot and it detects a rock in front of it. Instead of maybe turning 90º to the left or right it might go reverse then run into a wall and wonder why it can't keep going because it can't sense backwards.
b) To detect all 180º in front of it, we need 6 30º sensors.
c) A, C, D, F, H

Problem 4)

a) Torque: $T = T_s(1 - \frac{N_{load}}{N_{no\ load}}) \Rightarrow T_s = \frac{T}{(1 - N_{load}/N_{no\ load})} = \frac{0.5}{(1-250/1000)} = 0.67 kg\ cm$

b) $T_{new} = T_s \cdot gear\ ratio \Rightarrow 8 = 0.67 \cdot gear\ ratio \Rightarrow gear\ ratio = 12$

c) $N_{new} = \frac{N_{no\ load}}{gear\ ratio} = \frac{1000}{12} RPM,\ v = N_{new} \cdot 2\pi r = \frac{1000}{12} \cdot \pi \cdot 4 = 1047.20$ cm/min

Problem 5)

a) 7, shoulder has 3 (sideways, forward/back, rotate) + elbow has 1 (bending/straightening) + forearm has 1 (turn palm up/down) + wrist has 2 (sideways and up and down)

b) Robot has 2 cylindrical sections so that's + 2 DoF. But at point A it cements the human arm to the robot removing 6 DoF. As a result the total number of DoF is 7 from human + 2 from robot - 6 = 3 DoF. If I had to guess this would be to rehabilitate the arm since I had to do something similar when I tore my rotator cuff in my shoulder.

c)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	90^o	0	θ_2
3	0	0	d_3	0

Problem 6)

a) There are 3 joints so there are 3 DoF

b) 2 since L_2 and L_3 can go the full 360^o

c) $x = L_1 cos(\theta_1) + L_2 cos(\theta_2) + L_3 cos(\theta_3) \approx 0.94$

$y = L_1 sin(\theta_1) + L_2 sin(\theta_2) + L_3 sin(\theta_3) \approx 0.64$